KAPASA MAKASA UNIVERSITY

Department of Agriculture and Aquatic Sciences

Agribusiness Management (SA: 500)

Lecture 4: Economics for Agribusiness

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Introduction

- In the previous lecture, we looked at the basics of economics and how it relates to agribusiness management.
- In particular, we were able to differentiate and calculate the accounting profit and economic profit
- We also understand how supply and demand interact to determine market equilibrium
- In this lecture, we will focus on shifters of supply and demand, and end with elasticities of demand

Objectives

After finishing this lecture, students should be able to:

- Identify and describe what causes the entire supply and demand to shift
- Explain how agribusiness managers use price elasticity, income elasticity, and cross price elasticity of demand

The Economics of Markets

Shifts in Supply

A supply curve shifts due to determinants of supply:

- Change in technology
- Change in price of inputs
- Weather
- Price of related products (**Complements in Production** e.g., beef and leather & **Substitutes in Production** e.g., maize and soybeans).
- Subsidies or taxes
- Number of suppliers

The Economics of Markets

Shifters of Supply

- Supply depends on many variables:
- $Q_s = f(P, P_i, T, P_o, N, Other)$

Where:

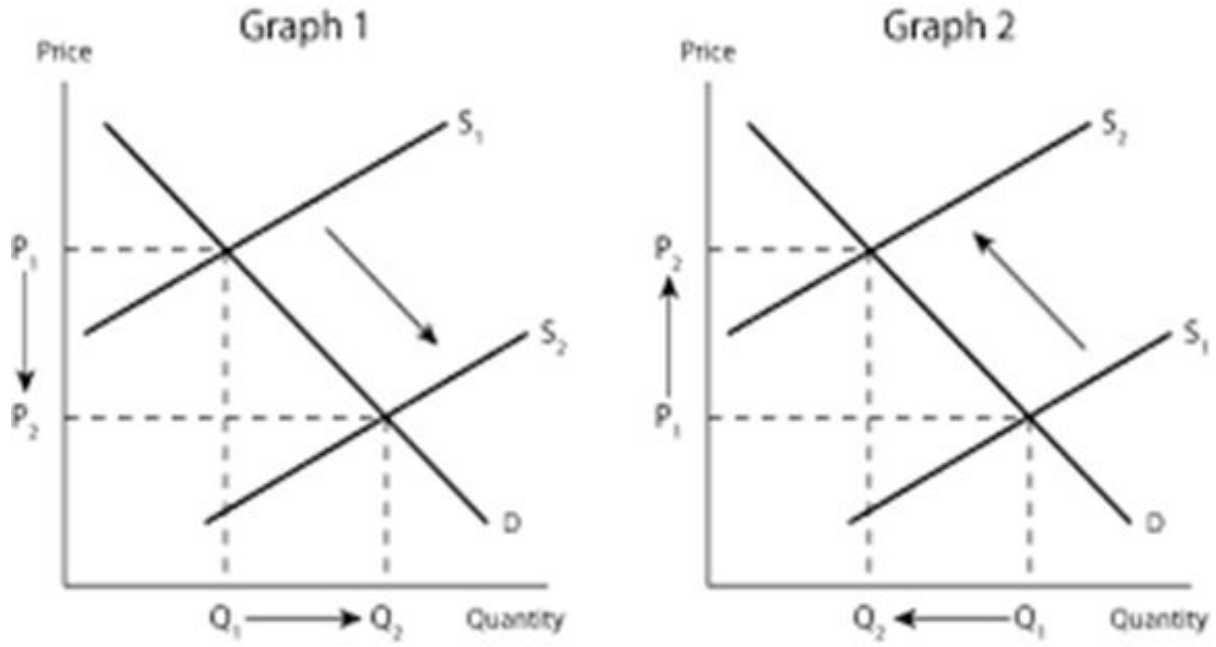
- P= own price
- P_i= input prices
- T= technology
- P_o= prices of related goods
- N= number of sellers

The Economics of Markets

Shifters of Supply

Example

- Diesel price increase, Supply shifts left
- Good rainfall, Supply shifts right



- Increase in input prices leads to increase in production costs & decreased supply → shift left.
- Decrease in input prices leads to costs fall & increased supply → shift right.

The Economics of Markets

Shifts in Demand

Five determinants of Shifts in demand:

- Income
- Tastes and preferences
- Expectations
- Number of buyers
- Price of substitutes or complements

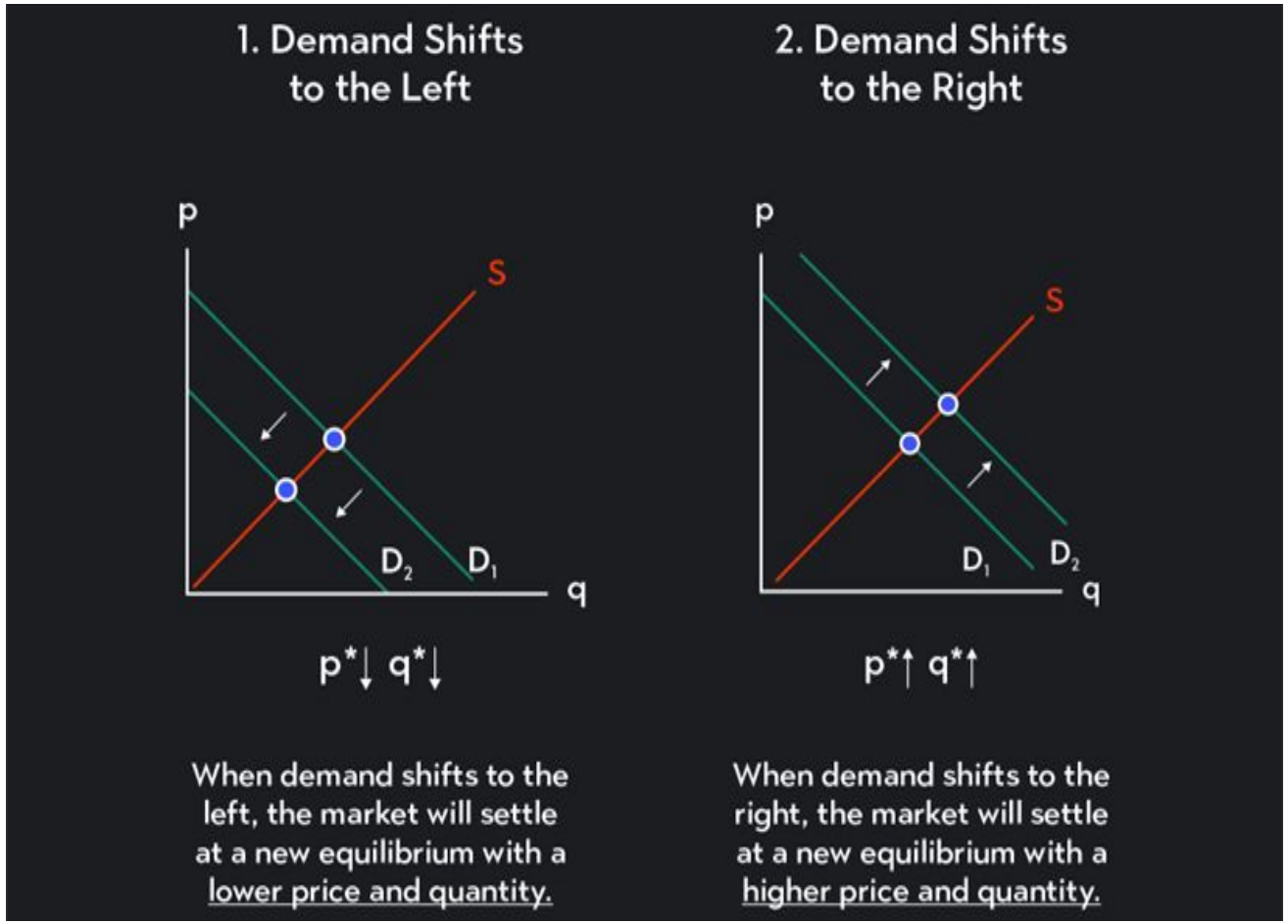
The Economics of Markets

Shifts in Demand

- Demand function:
- $Q_d = f(P_{own}, P_{related}, M, TP, EP, Pop)$Where:
- P_{own}= own price
- $P_{related}$= price of related goods
- M= income
- TP= tastes & preferences
- EP= expectations of future prices
- Pop= population

The Economics of Markets

Shifts in Demand



Interpretation:

- More quantity supplied at same price
- Curve shifts from S_0 to S_1
- This is a **change in demand**

Example

If income increases:

- Consumers buy more beef at every price.
- Demand shifts right.

The Economics of Markets

Derived Demand

- Derived demand is Demand for a product derived from demand for another product.

Example:

- Higher maize prices can lead to Higher demand for fertilizer

Elasticities of Demand

Elasticities of Demand is a measure of how quantity demanded responds to a change in **price**, **income**, **or price of other goods**.

Hence, three types of elasticities of demand:

(i) Price elasticity of demand: a measure of consumer response to price changes.

(ii) Income elasticity of demand: measures the response of quantity demanded to changes in income

(iii) Cross price elasticity of demand: measures how the quantity demanded for one product responds to a price change in a different product.

Elasticities of Demand

Importance of Elasticities of Demand to Agribusiness

- Elasticities of demand helps agribusiness managers to predict how consumers will respond to changes in **price**, **income** or **prices of other goods**.
- Understanding how to interpret the various types of elasticity allows agribusiness managers to make informed decisions.

(i) Calculating price elasticity of demand

- Price elasticity of demand reflects the percentage change in the quantity demanded when the price changes by 1 percent.
- If the quantity demanded for a bag of maize seed increases by 1 percent when its price decreases by 1 percent, the price elasticity of demand for maize seed is 1.0. The formula for price elasticity of demand or εd is:

$$\varepsilon_d = \frac{\%\ \text{change in quantity demanded}}{\%\ \text{change in price}} = \frac{(Q_2 \overrightarrow{\ } Q_1)}{(P_2 \overrightarrow{\ } P_1)} \times \frac{(P_1 + P_2)}{(Q_1 + Q_2)}$$

Elasticities of Demand

Interpretation of Price Elasticity of Demand

Absolute Value of Price Elasticity of Demand Coefficient	Demand is Price
$\lvert \varepsilon_d \rvert > 1.0$	Elastic
$\lvert \varepsilon_d \rvert = 1.0$	Unit elastic
$\lvert \varepsilon_d \rvert < 1.0$	Inelastic

Elasticities of Demand

- **Example 1:** Price Elasticity of Demand (εd)
- Price of a Bag of Rice falls from K400 to K300
- Quantity Demanded for increases from 100 to 200 units
- $\varepsilon_d = \frac{(Q_2 \rightarrow Q_1)}{(P_2 \rightarrow P_1)} \times \frac{(P_1 + P_2)}{(Q_1 + Q_2)}$
- $\varepsilon_d = \frac{(200 \rightarrow 100)}{(300 \rightarrow 400)} \times \frac{(400+300)}{(100+200)} = \frac{100}{-100} \times \frac{700}{300} = \frac{70{,}000}{-30{,}000} = \frac{70{,}000}{-30{,}000} = -2.33$
- $\varepsilon_d = -2.33$
- $|\text{ed}| = \mathbf{2.33}$
- **Since 2.33 > 1.0 Demand is elastic.**

Elasticities of Demand

Example 1: Price Elasticity of Demand (ɛd)

- This means that when the price drops by 1 percent, the quantity demanded increases by 2.33 percent.
- Because percentage change in quantity demanded shifted more than price shifted, the demand for Rice responded to the price change.

Example 2: What if the price fell from K200 to K100 per unit, and quantity demanded increased from 300 to 400?

Elasticities of Demand

Example 2: Price Elasticity of Demand (ɛd)

Step 1: We write the formula:

$$\varepsilon_d = \frac{(Q_2 \overrightarrow{\ } Q_1)}{(P_2 \overrightarrow{\ } P_1)} \times \frac{(P_1 + P_2)}{(Q_1 + Q_2)}$$

Step 2: Identify Requirements:

$Q_2 = Quantity\ 2 = 400$

$Q_1 = Quantity\ 2 = 300$

$P_1 = Price\ 1 = K200$

$P_2 = Price\ 2 = K100$

Step 3: We place the figures in the formula

$$\varepsilon_d = \frac{400 - 300}{200 - 100} \times \frac{200 + 100}{400 + 300}$$

$$= \frac{100}{-100} \times \frac{300}{700} = -1 \times \frac{3}{7} = -0.43$$

Since we are only interested in the magnitude of change in demand, we ignore the sign of the number by introducing the absolute value sign **|** **|** i.e. we take the number without its sign

$|\text{ed}| = \mathbf{0.43}$

Since $\mathbf{0.43 < 1}$ **Demand is inelastic**

Elasticities of Demand

- The coefficient for price elasticity of demand should always be negative, since change is measured along the demand curve which has a negative slope.
- As price increases, quantity demanded decreases, or vice versa.
- So, calculated price elasticities of demand will always be negative.
- Typically, one takes the absolute value of price elasticity of demand, and thus, ignores the negative sign.

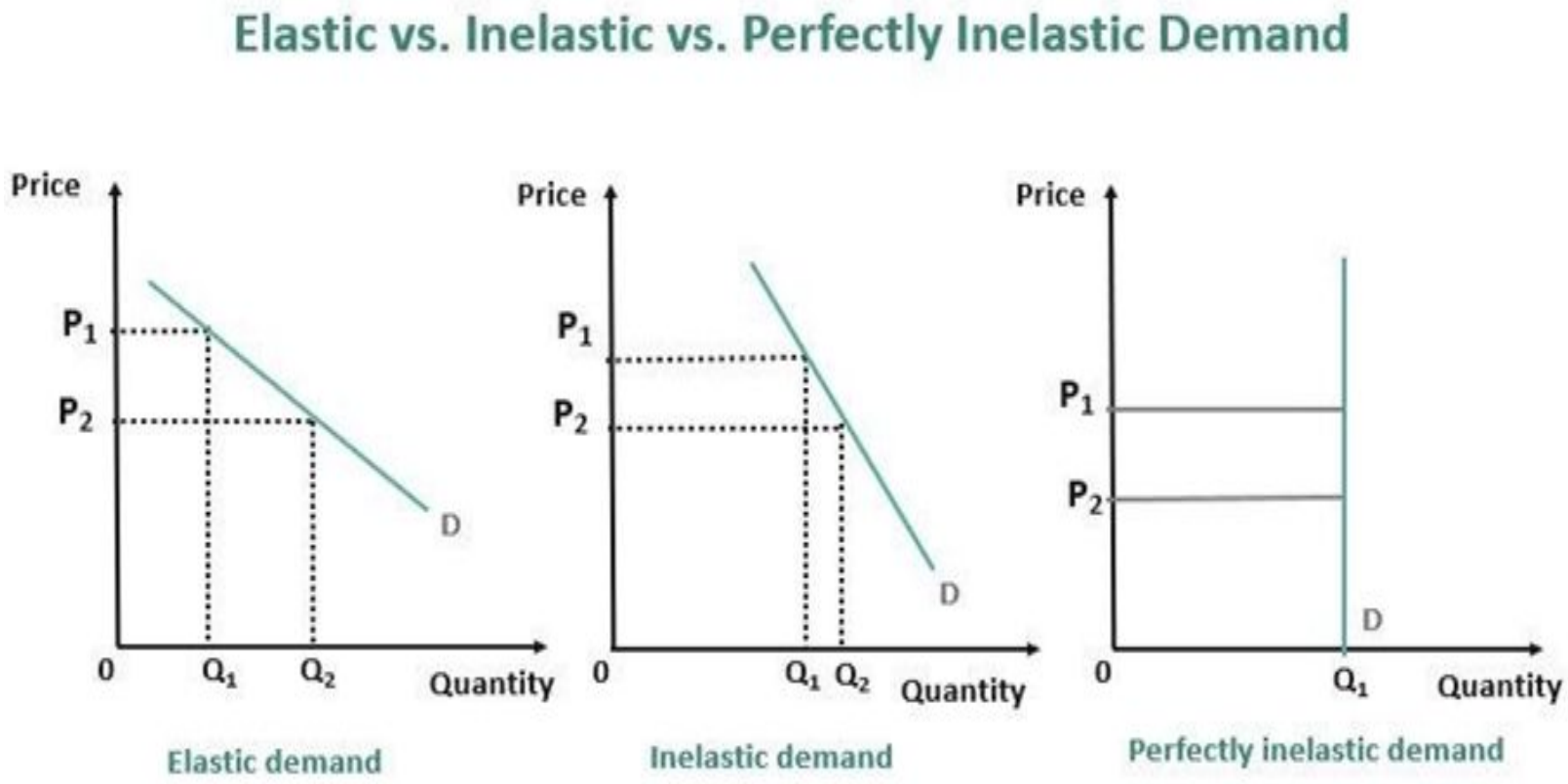
Elasticities of Demand

Inelastic Demand

Inelastic Demand is when a change in price brings about a relatively smaller change in quantity demanded.

Examples:

- Maize meal
- Fertilizer (short run)
- Fuel
- Medicines

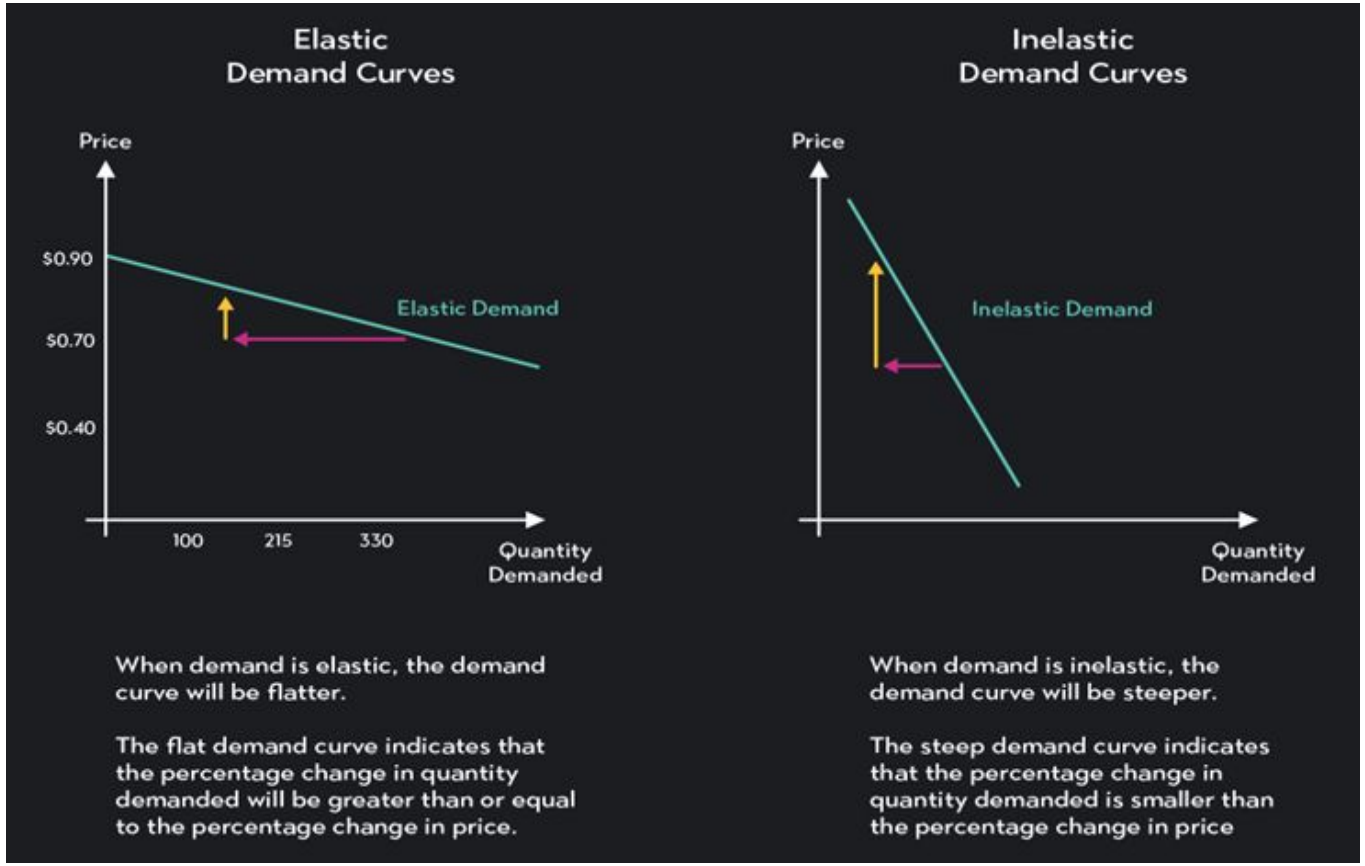


Elasticities of Demand

- **ELASTIC DEMAND**
- **Elastic Demand** is when a change in price brings about a relatively larger change in quantity demanded.

Examples:

- Avocados
- Apples
- Restaurant meals
- Imported fruits



Elasticities of Demand

What Determines Price Elasticity of Demand?

(i) Availability of Substitutes

- More substitutes, More elastic
- Fewer substitutes, More inelastic

Examples

Product	Elasticity	Reason
Maize (staple)	Inelastic	Few close substitutes
Beef	Elastic	Chicken & fish available
One brand of cooking oil	Very elastic	Many brands available
All cooking oil	Less elastic	Few alternatives

Elasticities Of Demand

(ii) ELASTICITY & TIME

- Short run, More inelastic
- Long run, More elastic

Example:

If electricity price increased:

- Short run: demand for electricity cannot change: inelastic
- Long run: install solar: elastic

Elasticities of Demand

(iii) ELASTICITY & TOTAL REVENUE

$$TR = P \times Q$$

Rule

Demand Type	If Price Increased	Total Revenue
Inelastic	Revenue Increases	Increased
Elastic	Revenue Decreases	Decreased
Unitary	Revenue Unchanged	Unchanged

If maize demand is inelastic:

- Price increase, total revenue rises.

If avocado demand is elastic:

- Price increase, total revenue falls.

Elasticities of Demand

(ii) Cross-price Elasticity Of Demand

- Used to classify how demand relates to price changes in other goods.

The relationship between two goods can be classified as:

- **Complementary goods**
- **Independent goods**
- **Substitute goods**

Grocers, for example, carefully study these relationships as they determine how to adjust orders of different items as they put certain goods on sale each week

- Cross price elasticity of demand < 0 Complement goods
- Cross price elasticity of demand = 0 Independent goods
- Cross price elasticity of demand > 0 Substitute goods

Elasticities of Demand

(ii) Cross-price Elasticity Of Demand

- If the sign for cross price elasticity of demand is negative, the two goods are deemed to be complements, like Sugar and buns.
- This means if the price of buns is reduced, consumers will also buy complementary goods that are eaten with the buns such as sugar.
- If the sign for cross price elasticity of demand is positive, the goods are substitutes for each other.
- One classic example is Coke and Pepsi. If the price of Pepsi is higher, the demand for Coke will increase.
- If the sign is near zero, it means that the demand for one good does not depend on the price change in another good. For example, the demand for coffee will not change if the price of Cooking oil is reduced.

Elasticities of Demand

(ii) Cross-price Elasticity Of Demand

(i) Substitutes

Examples:

- Butter & margarine
- Beef & chicken
- If Butter price rises, margarine demand rises.
- Cross elasticity is **positive**.

Elasticities of Demand

(Ii) COMPLEMENTS

Goods consumed together.

Examples:

- Bread & butter
- Fertilizer & hybrid seeds
- Tractor & diesel
- If tractor price increases, diesel demand decreases.
- Cross elasticity is **negative**.

Elasticities of Demand

Income Elasticity Of Demand

- **Income Elasticity of Demand (YED)** measures the responsiveness of quantity demanded to a change in consumer income.
- It shows how demand changes when **income changes**, holding prices constant.
- **Formula**
- $YED = \frac{\%\ \Delta Q_d}{\%\ \Delta Y}$

Where:

- $\%\ \Delta Q_d$= Percentage change in quantity demanded
- $\%\ \Delta Y$= Percentage change in income
- $YED = \frac{(Q_2 \rightarrow Q_1)}{(Y_2 \rightarrow Y_1)} \times \frac{(Y_1 + Y_2)}{(Q_1 + Q_2)}$

Elasticities of Demand

Types Of Goods Based On Income Elasticity

1. Normal Goods

- Goods for which demand increases when income increases, prices held constant.
- $YED > 0$

Examples:

- Milk
- Beef
- Fruits
- Eggs
- Hybrid seed (commercial farmers)

Elasticities of Demand

Types Of Goods Based On Income Elasticity

- **2. Inferior Goods**
- Goods for which demand decreases when income increases.
- $YED < 0$

Examples:

- Very low-quality maize meal
- Broken rice
- Second-hand clothing
- Cassava in some urban areas

As income rises, consumers switch to better-quality alternatives.

Elasticities of Demand

Types Of Goods Based On Income Elasticity

3. Luxury Goods

- Goods for which demand increases more than proportionately as income increases.
- $YED > 1$

Examples:

- Organic foods
- Imported fruits
- Restaurant meals
- Premium beef cuts

Elasticities of Demand

Types Of Goods Based On Income Elasticity

4. Necessities

- Goods for which demand increases less than proportionately as income increases.
- $0 < YED < 1$

Examples:

- Staple maize meal
- Salt
- Cooking oil
- Basic vegetables

Elasticities of Demand

Types Of Goods Based On Income Elasticity

Type of Good	YED Value	Example
Inferior	< 0	Low-grade maize
Necessity	0–1	Staple maize meal
Luxury	> 1	Organic food
Normal	> 0	Milk, eggs

Elasticities of Demand

EXAMPLE of Income Elasticity

Income rises from K2,000 to K2,500 per month.

Demand for beef increases from 10 kg to 14 kg.

Step 1: Calculate % change in Quantity

$$\%\Delta Q = \frac{14 - 10}{10} \times 100 = 40\%$$

Step 2: Calculate % change in Income

$$\%\Delta Y = \frac{2500 - 2000}{2000} \times 100 = 25\%$$

Step

3: Compute Income Elasticity

$$YED = \frac{40\%}{25\%} = 1.6$$

Interpretation:

Since $YED = 1.6 > 1$

Beef is a **luxury good** in this example.

Elasticities of Demand

AGRIBUSINESS APPLICATIONS OF INCOME ELASTICITY

- Forecasting Demand
- Business planning
- Market Targeting

The End